

curves and other geometric shapes and then stored in image files. In this chapter, we start with drawing tools, move on to vector graphic essentials and then finally see some tools for creating and editing vector graphics.

4.1. Graphics Tablets

A graphics tablet (or digitising tablet, graphics pad, drawing tablet) is a computer input device that allows you to hand-draw images and graphics, similar to sketching with pencil and paper.

A graphics tablet consists of a flat surface upon which the user may “draw” an image using an attached stylus, a pen-like drawing apparatus. The image generally does not appear on the tablet itself, but is rather displayed on the computer monitor.

Some tablets are intended as a replacement for the mouse as the primary pointing and navigation device for desktop computers.

Early graphics tablets are most commonly known as spark or acoustic tablets and are used as a stylus that generated clicks with a spark plug. The clicks were then triangulated by a series of microphones to locate the pen in space. The system was fairly complex and expensive, and the sensors were susceptible to interference by external noise.

Types Of Graphics Tablets

Passive tablets

Passive tablets, most notably those by Wacom, make use of electromagnetic induction technology, where the horizontal and vertical wires of the tablet operate as both transmitting and receiving coils (as opposed to the wires of the RAND Tablet which only transmit). The tablet generates an electromagnetic signal, which is received by the LC circuit in the pen. The wires in the tablet then change to a receiving mode and read the signal generated by the pen. Modern arrangements also provide pressure sensitivity and one or more switches (similar to the buttons on a mouse), with the electronics